Speed, Distance & Time

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: A cyclist travels 20 km in 1 hour. What is her speed?

Answer: Speed = distance ÷ time = 20 ÷ 1 = 20 km/h.

Standard: A car travels at 30 km/h for 3 hours. How far does it travel?

Answer: Distance = speed × time = 30 × 3 = 90 km.

Advanced: A train covers 150 km in 3 hours. What is its average speed?

Answer: Speed = distance ÷ time = 150 ÷ 3 = 50 km/h.

Easy / Standard (15 questions)

1. A car travels 60 km in 1 hour. What is its speed?
2. Which formula do we use to find distance travelled? A) Speed × Time B) Speed ÷ Time C) Speed − Time D) Time ÷ Speed
3. A cyclist rides at 20 km/h for 3 hours. How far does she travel?
4. A plane flies 400 km in 2 hours. What is its speed?
5. Which unit correctly measures speed? A) km B) hours C) km/h D) kg
6. A walker moves at 5 km/h for 4 hours. How far does she walk?
7. A car travels 90 km in 3 hours. What is its average speed?
8. A train travels at 80 km/h for 2 hours. How far does it travel?
9. Select ALL of the speed calculations below that are correct: A) 100 km in 2 hours = 50 km/h B) 60 km in 3 hours = 30 km/h C) 90 km in 3 hours = 30 km/h D) 40 km in 4 hours = 10 km/h E) 24 km in 2 hours = 8 km/h
10. A car travels 150 km in 3 hours. What is its speed?
11. A runner covers 8 km in 2 hours. What is her speed?
12. Which of these speeds is the fastest? A) 30 km/h B) 45 km/h C) 25 km/h D) 40 km/h
13. A coach travels at 50 km/h for 2 hours. How far does it travel?
14. A car takes 4 hours to travel 200 km. What is its average speed?
15. A cyclist travels 12 km in the first hour and 12 km in the second hour. What is his average speed over the 2 hours?

Standard (15 questions)

1. A ship sails 240 km in 4 hours. What is its speed?
2. A tractor drives at 15 km/h for 4 hours. How far does it travel?
3. A car travels 180 km in 3 hours. What is its speed? A) 50 km/h B) 60 km/h C) 70 km/h D) 90 km/h

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4. A horse trots at 12 km/h for 3 hours. How far does it travel?

5. A train travels 210 km in 3 hours. What is its speed?

6. A bus travels at 40 km/h for 5 hours. How far does it travel?

7. A plane flies at 600 km/h for 2 hours. How far does it travel? A) 300 km B) 900 km C) 1,200 km D) 1,600 km

8. A cyclist covers 45 km in 3 hours. What is her speed?

9. A car travels 320 km in 4 hours. What is its speed?

10. A walker travels at 6 km/h for 5 hours. How far does he walk?

11. Select ALL statements that are TRUE about a car travelling at 50 km/h: A) In 2 hours it travels 100 km B) In 3 hours it travels 200 km C) In 1 hour it travels 50 km D) In 4 hours it travels 300 km E) In 5 hours it travels 250 km

12. A train takes 5 hours to travel 350 km. What is its average speed?

13. A coach travels at 45 km/h for 4 hours. How far does it travel?

14. Which journey has the greatest speed? A) 120 km in 3 hours B) 100 km in 2 hours C) 180 km in 4 hours D) 90 km in 3 hours

15. A car travels 150 km in the first 3 hours and 150 km in the next 3 hours. What is its average speed for the whole journey?

Standard / Hard (15 questions)

1. On a long drive, a family covers 100 km during the first 2 hours, then a further 60 km over the next 2 hours. Work out their average speed for the whole drive.

2. Travelling at a steady 60 km/h, a train completes a 180 km route. How many hours does the trip take? A) 2 hours B) 3 hours C) 4 hours D) 5 hours

3. Maya cycles 20 km over the first 2 hours of a trip, then covers another 10 km in the following hour. What is her average speed across the whole trip?

4. Flying at a constant 200 km/h, an aeroplane needs to cover 600 km. How many hours will the flight last?

5. A delivery van drives at 40 km/h for the first 2 hours of its route, then speeds up to 50 km/h for 1 more hour. What total distance has it covered?

6. On the way out, a coach covers 180 km in 3 hours. Coming back along the same 180 km route takes it an extra hour. What speed does it average on the return leg? A) 40 km/h B) 45 km/h C) 50 km/h D) 60 km/h

7. A cargo ship holds a speed of 20 km/h for its first 3 hours at sea, then increases to 30 km/h for the next 2 hours. What total distance has it sailed?

8. Starting from the same spot and at the same moment, two cars drive off in opposite directions — one at 30 km/h and the other at 50 km/h. After 2 hours, what distance separates them?

9. A 200 km road trip is driven at a steady average of 40 km/h throughout. Select ALL of the statements below that are TRUE about this trip: A) It takes 5 hours altogether B) It takes 4 hours altogether C) After 2 hours, 80 km has been covered D) Covering half the trip (100 km) takes 2.5 hours E) The car covers 40 km in every hour of driving

10. A 240 km trip is split into two equal halves: the car drives the first half at 60 km/h and the second half at 40 km/h. How many hours does the complete trip take?

11. Setting off at 20 km/h, a cyclist rides 40 km, stops to rest for a full hour, then continues for a further 20 km at a slower 10 km/h. Counting the rest stop, how long does the entire trip take?

12. Departing at 09:00 and holding a constant 80 km/h, how far will a train have travelled by the time the clock reaches 11:00? A) 80 km B) 120 km C) 160 km D) 200 km

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13. The first stage of a car's journey covers 180 km in 3 hours; the second stage adds a further 60 km in just 1 hour. Find the average speed across the whole journey.

14. A cyclist and a runner start from the same spot at the same moment and head off in the same direction, the cyclist moving at 20 km/h and the runner at 8 km/h. What gap has opened up between them after 2 hours?

15. A lorry's 300 km delivery route begins with 180 km driven at 60 km/h. To finish the entire route in exactly 5 hours, what speed is needed for the final 120 km?

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